Machine Learning-Based Analysis of Household Expenditure

Urban Households of Maharashtra

1. Project Overview

This project investigates household expenditure patterns in urban Maharashtra using machine learning techniques. It focuses on predicting monthly transportation and food expenditure using socioeconomic and demographic characteristics.

2. Dataset Information

Parameter	Details
Total observations	10,567 households
Training observations	8,453 (80%)
Testing observations	2,114 (20%)
Numerical feature	Total household income
Categorical features	District, age, occupation, education, gender, household size
Prediction targets	Monthly transportation and food expenditure

3. Machine Learning Models

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor
4. Gradient Boosting Regressor
5. Bootstrap Aggregation (Bagging Regressor)
6. Support Vector Regression (SVR)

4. Model Performance Comparison

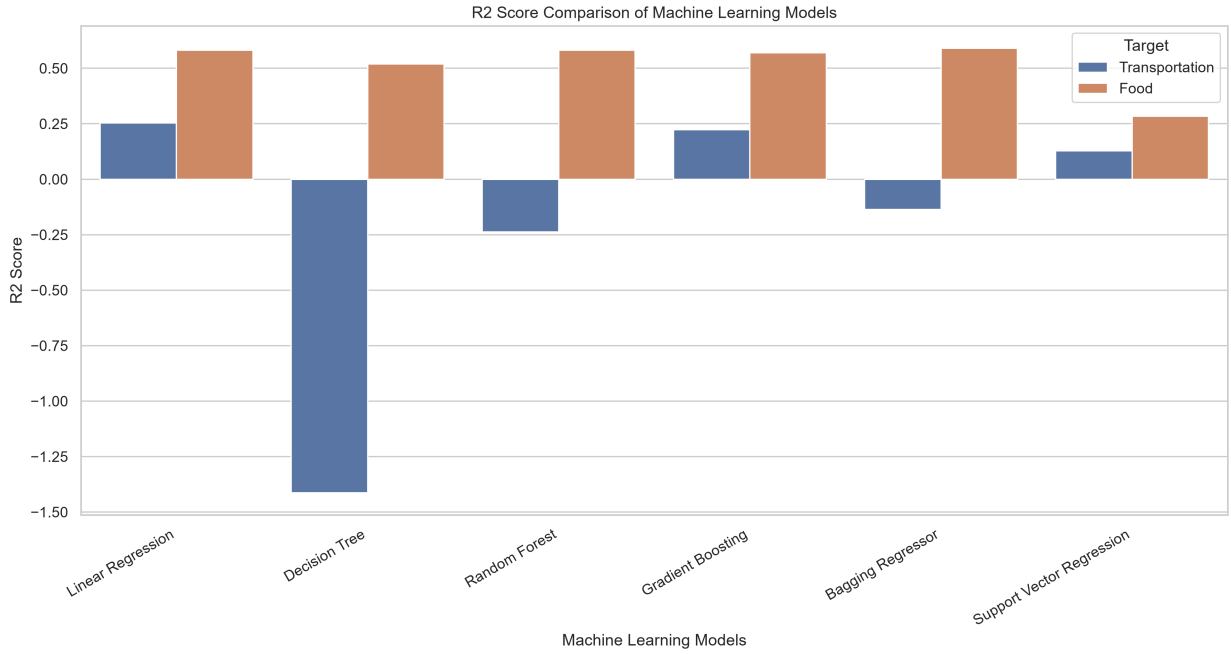
Target	Model	R^2	MAE	RMSE
Food	Bagging Regressor	0.5904	1782.69	2621.88
Food	Linear Regression	0.5820	1863.88	2648.79
Food	Random Forest	0.5812	1801.99	2651.17
Food	Gradient Boosting	0.5694	1882.47	2688.25
Food	Decision Tree	0.5189	1921.58	2841.51
Food	Support Vector Regression	0.2836	2377.20	3467.58
Transportation	Linear Regression	0.2542	526.28	1098.36
Transportation	Gradient Boosting	0.2231	529.07	1120.98
Transportation	Support Vector Regression	0.1285	512.54	1187.30
Transportation	Bagging Regressor	-0.1369	576.18	1356.11
Transportation	Random Forest	-0.2379	585.99	1415.03
Transportation	Decision Tree	-1.4132	631.65	1975.71

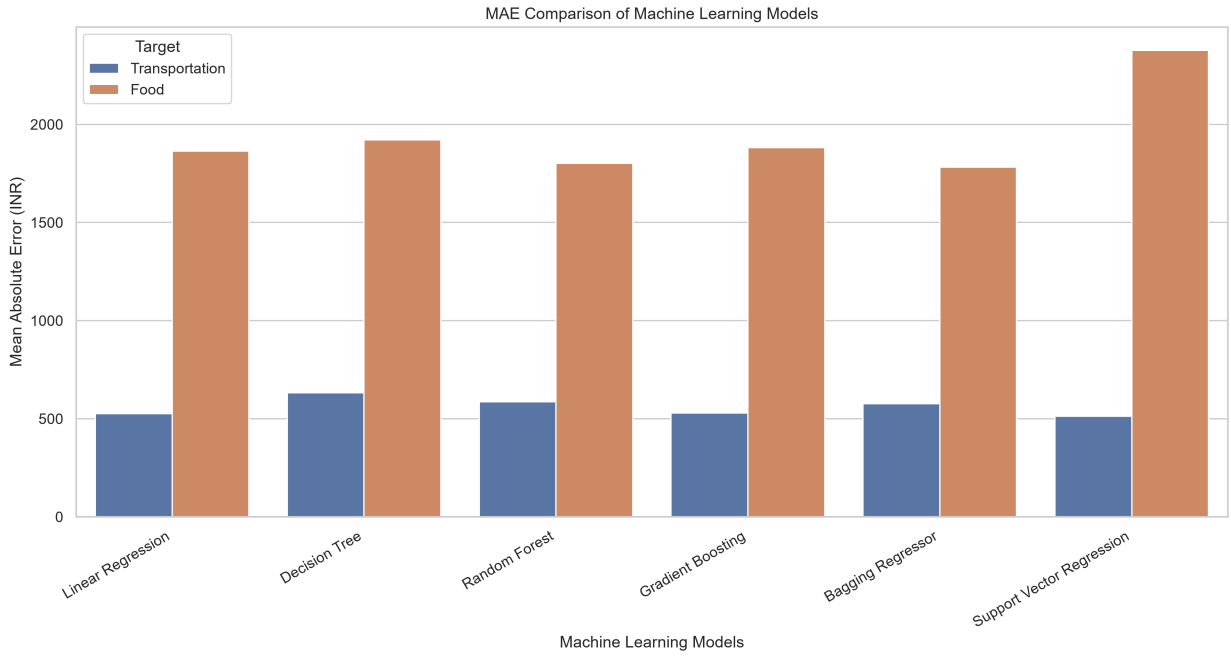
5. Best Performing Models

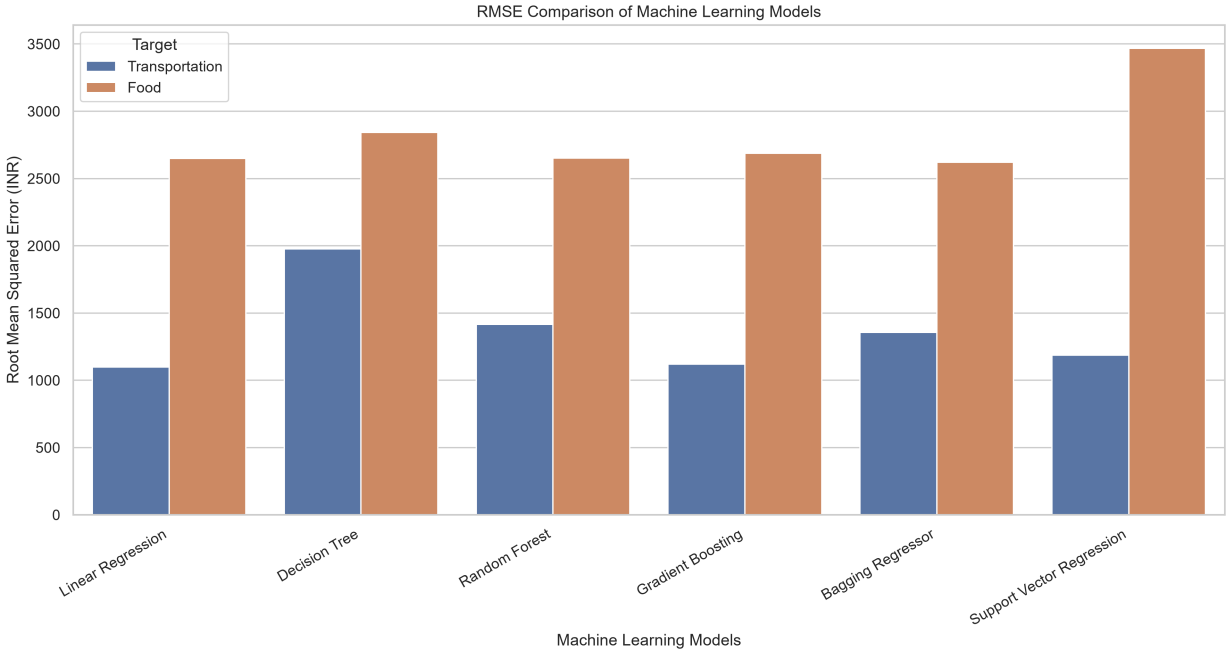
Transportation:
Best R^2 model: Linear Regression (R^2 = 0.2542)
Best MAE model: Support Vector Regression (MAE = 512.54)
Best RMSE model: Linear Regression (RMSE = 1098.36)

Food:
Best R^2 model: Bagging Regressor (R^2 = 0.5904)
Best MAE model: Bagging Regressor (MAE = 1782.69)
Best RMSE model: Bagging Regressor (RMSE = 2621.88)

6. Visual Model Comparison







7. Conclusion

The six machine learning algorithms were evaluated for predicting monthly household transportation and food expenditure. R^2, MAE and RMSE were used to compare predictive performance. The best-performing model may differ between the two expenditure categories. These results represent predictive associations and should not be interpreted as evidence of causation.